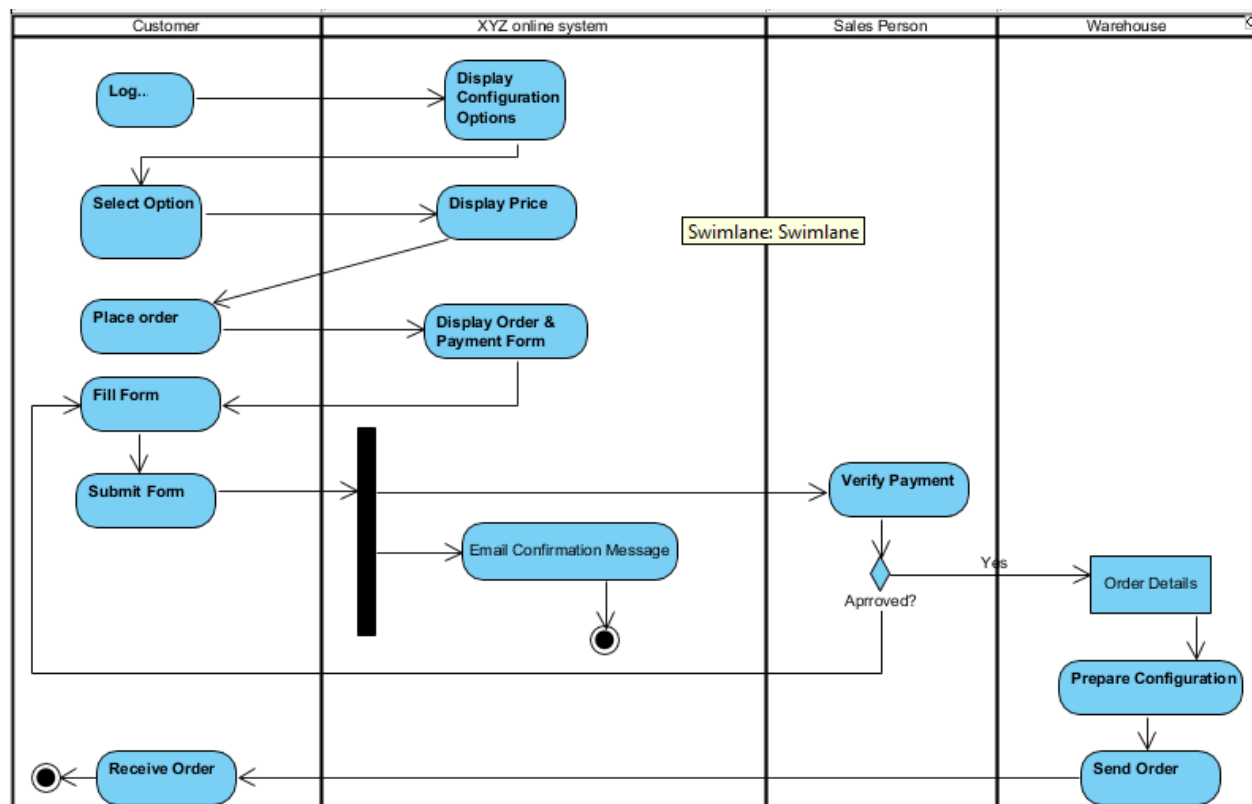


Tutorial 9 – Software Design Notations

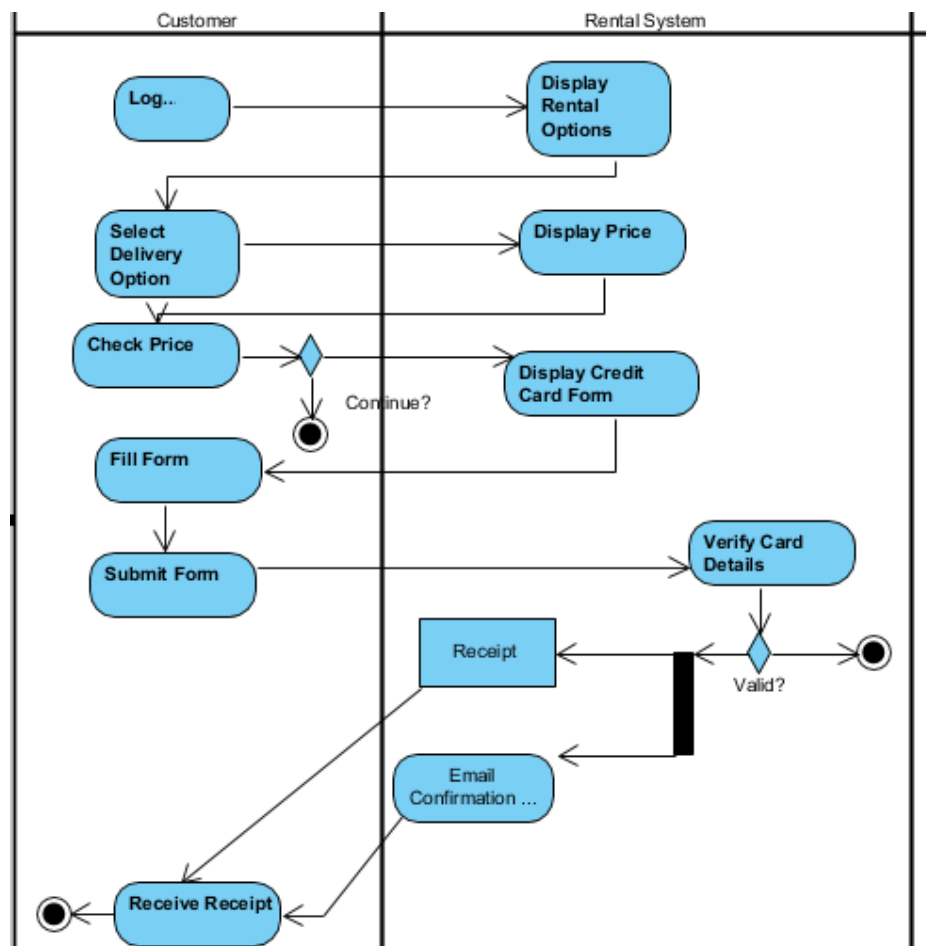
- XYZ is a company which sells computer via online. When a customer wishes to purchase computers from XYZ, he/she is required to log on to the company's website. The online system will then display the webpage with different sets of standard computer configurations options. The customer will select to view a particular set of standard configuration option. The price is then shown by the system automatically. The customer may then place the order by clicking on the order button. The system will then display an online order form with shipment and the payment details to be filled up by the customer. Once the order details have been submitted by the customer, the system will send an email confirmation message to the customer with details of the order. The salesperson at the sales department of the company will verify the customer's credentials and payment method. Once the approval has been made, the salesperson sends a request to the warehouse with details of the order. The warehouse staff will prepare the computer configuration ordered by the customer and deliver it to the customer's address.

Draw an **activity diagram** for the processing of the online shopping customer order as described above. Include the swimlanes and object flows in your diagram.



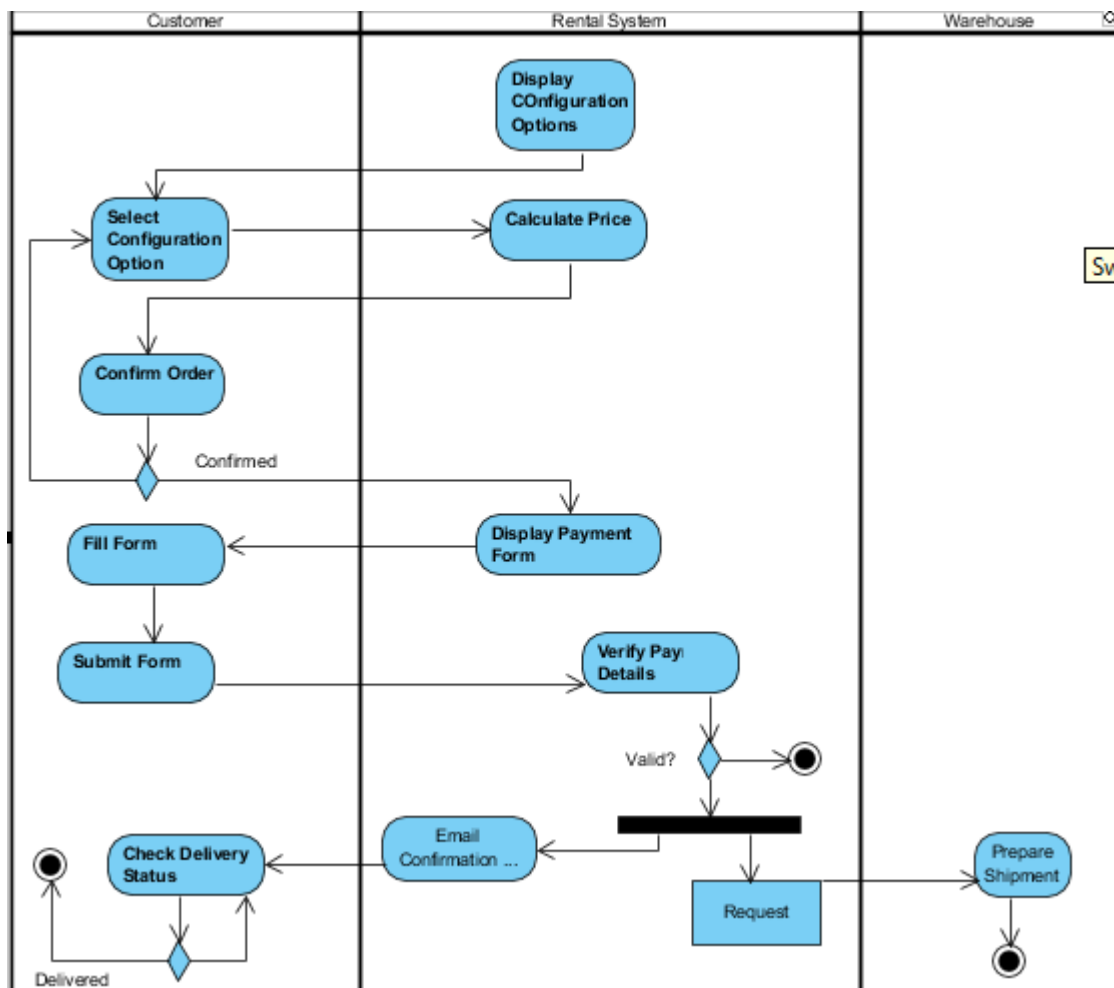
2. CarRent is a car rental company in Malaysia which provides an online service for car rental. When a customer wishes to rent a car from CarRent, he/she is required to login to the company's website. The website will then display the rental options where the customer has to select pick up location, date and time and also the vehicle type. The price and terms is shown by the system automatically. The customer may continue with the rental of the car by clicking continue button. The system will then display car rental information with a credit card form to be filled up by the customer. After the credit card details is submitted by the customer, the system will verify the credit card details. Once the credit card is verified, the system will automatically send an email confirmation message to the customer and generate a receipt with car rental details for the customer.

Draw an **activity diagram** for the **Car Rental System** as described above. Include swimlanes, synchronization bars and object flows in your diagram.



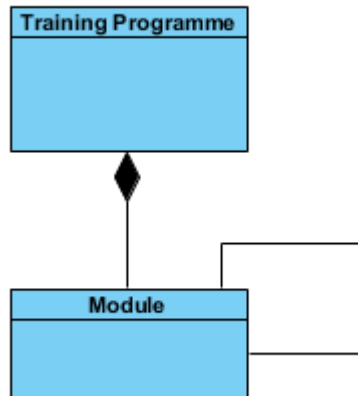
3. EZCom is a computer company established for more than three years. Recently the company is planning to offer the possibility of purchasing computers through an online system to their customers. The system will allow the customers to select a standard configuration or build a desired configuration for their computers online. The computers are classified as servers, desktop and portables. For each new configuration (such as memory), the system will calculate the price. To place an order, a customer must fill out the shipment and payment information (through credit card). Once the customer's credentials and payment is verified, the system will send request to warehouse about the ordered configuration to be shipped to the customer. At the same time, the system sends confirmation email messages to the customer with details of his/her order. While waiting for the arrival of the computer, the customer can check the status of his/her order online at any time.

Construct an **activity diagram** for the online system above. Your diagram must include relevant swimlanes, synchronization bars and object flows.



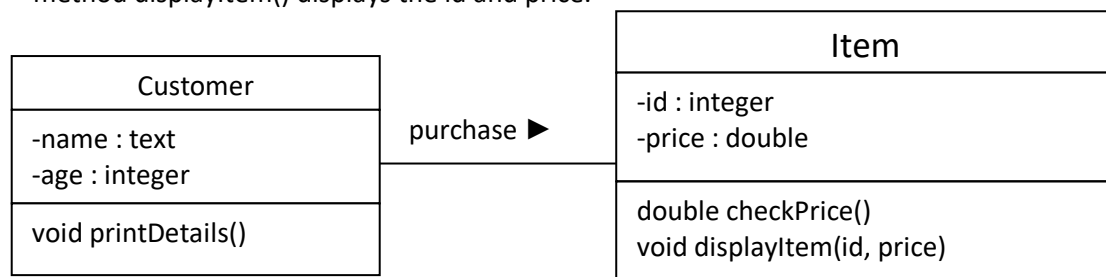
4. Draw a class diagram based on the description as described below:

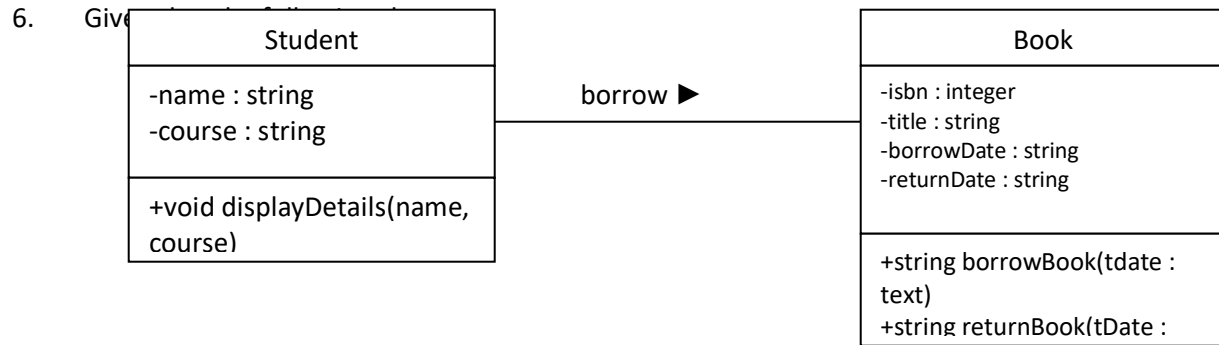
A software training program is identified by a code and has a short description that explains its goals. Each software training program is divided into modules, and each module is described by module name and the appropriate practice time. Each module sometimes has prerequisite modules.



5. Redraw the following class diagram by adding the given details:

- All the attributes are private visible and all methods are public visible.
- In Customer class, set the name attribute to text type, age attribute to integer type. The method printDetails() just displays the attributes' values.
- In Item class, set the id attribute to integer type, and price attribute to store double-precision, floating-point number. The method checkPrice() returns the price of the item, and method displayItem() displays the id and price.



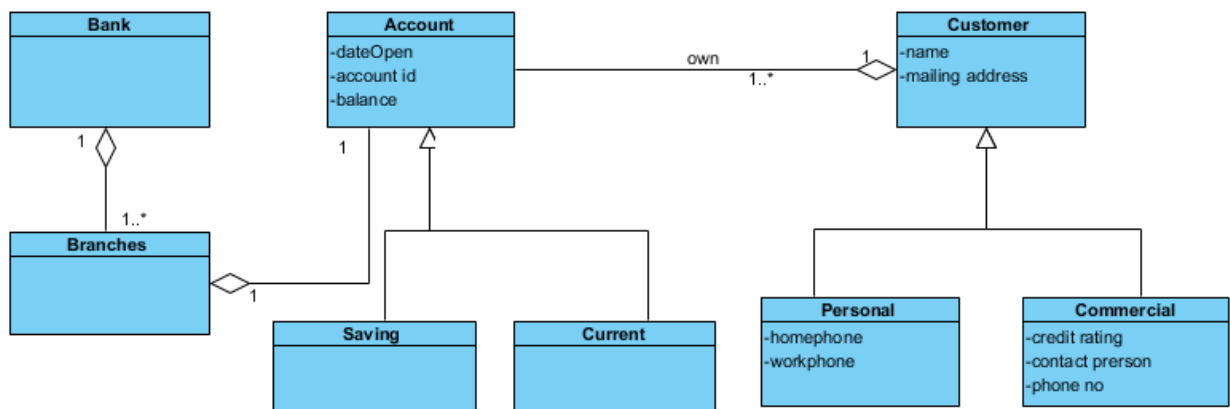


Redraw above class diagram by adding the following details:

- All attributes are having private visibilities and all methods are having public visibilities.
- Under student class, set the name and course attribute to text data type. The displayDetails() method prints all the student's attributes on the screen.
- Under book class, set the isbn to integer data type, whereas set the title, borrow date and return date to text data type. The borrowBook() method accepts a date of type text and set the value to the borrowDate attribute. The returnBook() method accepts a date of type text and set the value to the returnDate attribute.

7. Draw a class diagram based on the bank description as described below:

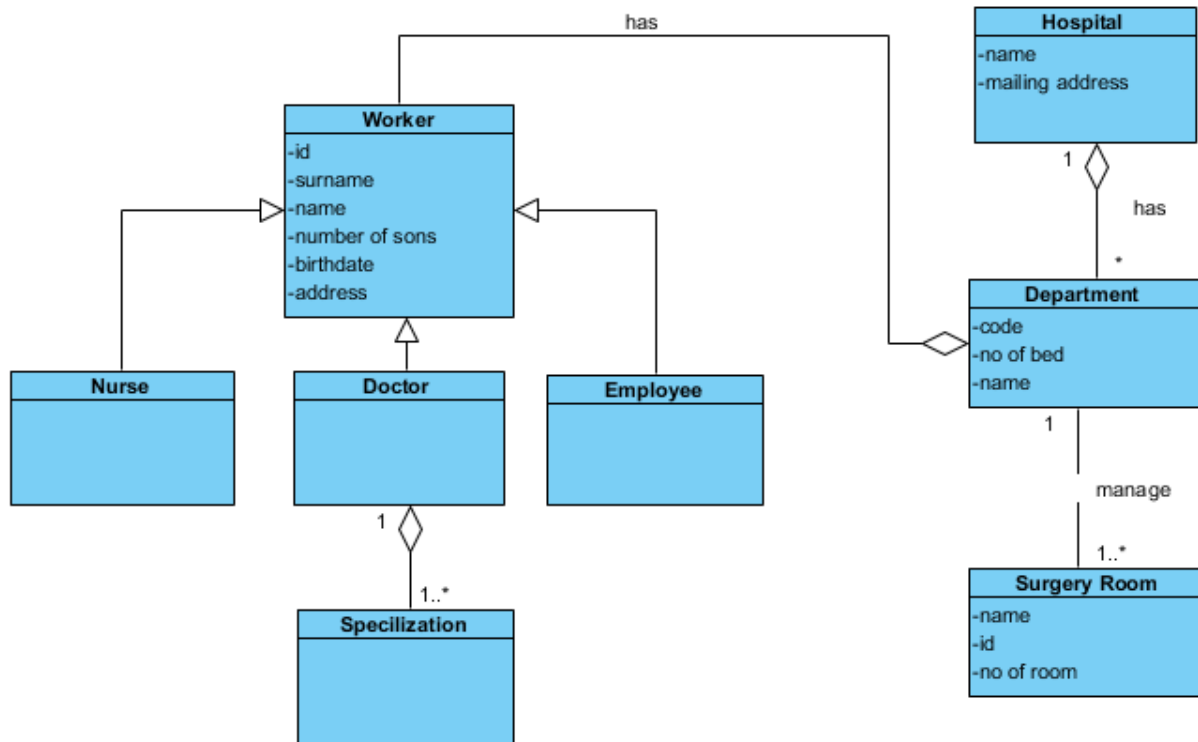
There are two special types of customers – personal and commercial. All customers have a name and mailing address. Commercial customers have additional attributes for credit rating, contact person, and contact person phone. Personal customers have attributes for home phone and work phone. The bank has multiple branches, and each account is serviced by one branch. Naturally, each branch has many accounts. Each customer can have more than one account but each account is uniquely owned by a customer. There are two types of accounts namely Savings and Current account. The accounts have information such as date opened, account number and balance.



*****Attributes should be allocated to the classes if possible

8. Draw a class diagram based on the following scenario:

Hospitals have an ID, a name and an address. People working in a hospital have an ID (unique in the hospital), surname, name, birth date, address, and number of sons. Hospital workers can be doctors (having a list of specializations), employees (having a role) and nurses. Hospital is divided into departments. Each department has a code, a name, and a number of beds. For each department, we know the primary doctor and all the workers. Each department can manage surgery rooms identified by name, ID and number of surgery tables.

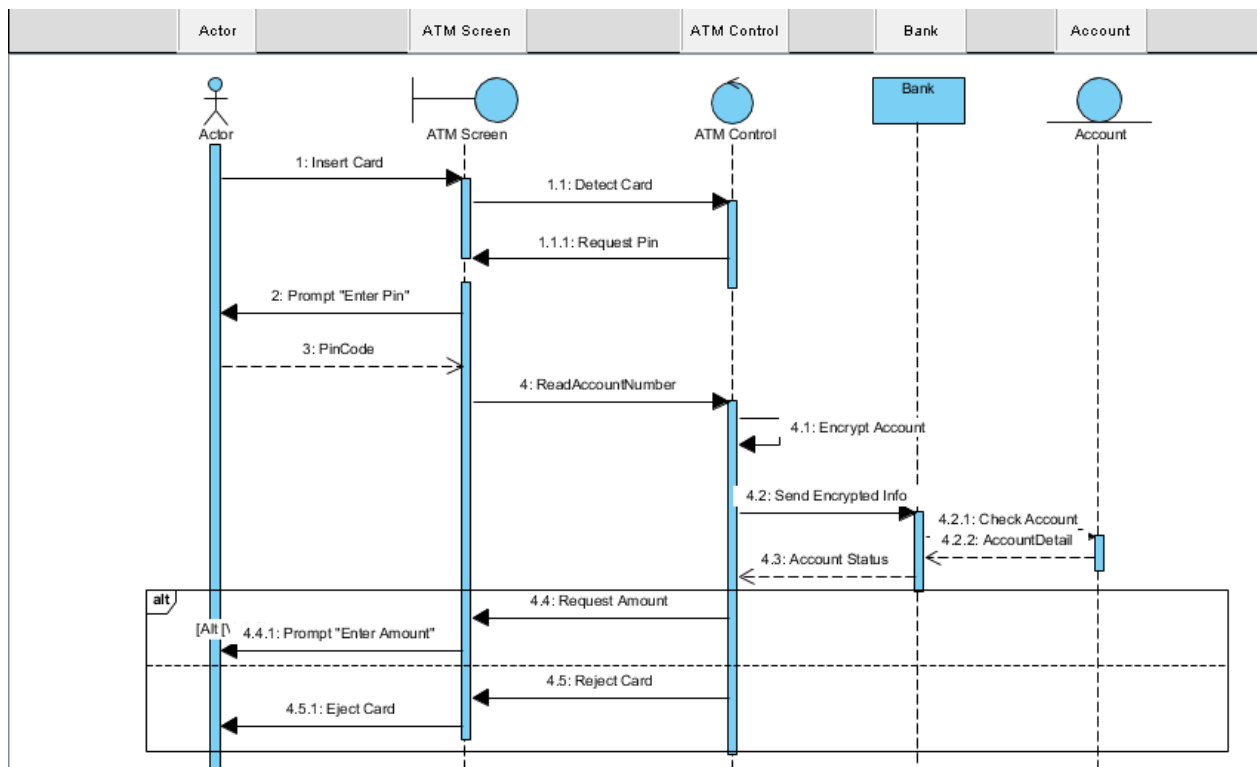


***** You may put a **class ROLE** with composition/aggregation to employee

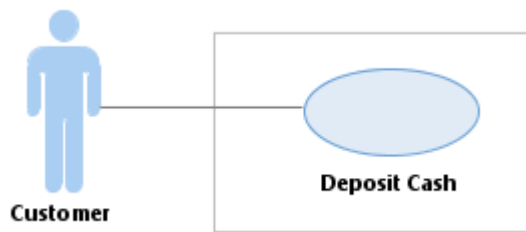
9. Given the scenario below, construct a **design sequence/ collaboration diagram** that shows the normal flow or events for the use case “withdraw money”.

A card holder wants to withdraw money from his bank account. He inserts his card into an ATM (automatic teller machine). The ATM prompts “Enter PIN”, and the card holder enters his PIN. The ATM internally retrieves the bank account number from the card. The ATM encrypts the PIN and the account number and sends it over to the bank. The bank verifies the encrypted account number and PIN. If the PIN is correct, the ATM displays “Enter amount” to allow the card holder to withdraw money and the ATM pays out the money.

You are required to include boundary (UI), control, entity objects (account and customer) and object creation in your diagram.



10. Construct a **design sequence/ collaboration diagram** that shows the normal flow of events for the use case “deposit cash” which carry out by customer to deposit cash to an account.



You are required to include boundary (UI), control, entity objects (account and customer) and object creation in your diagram.

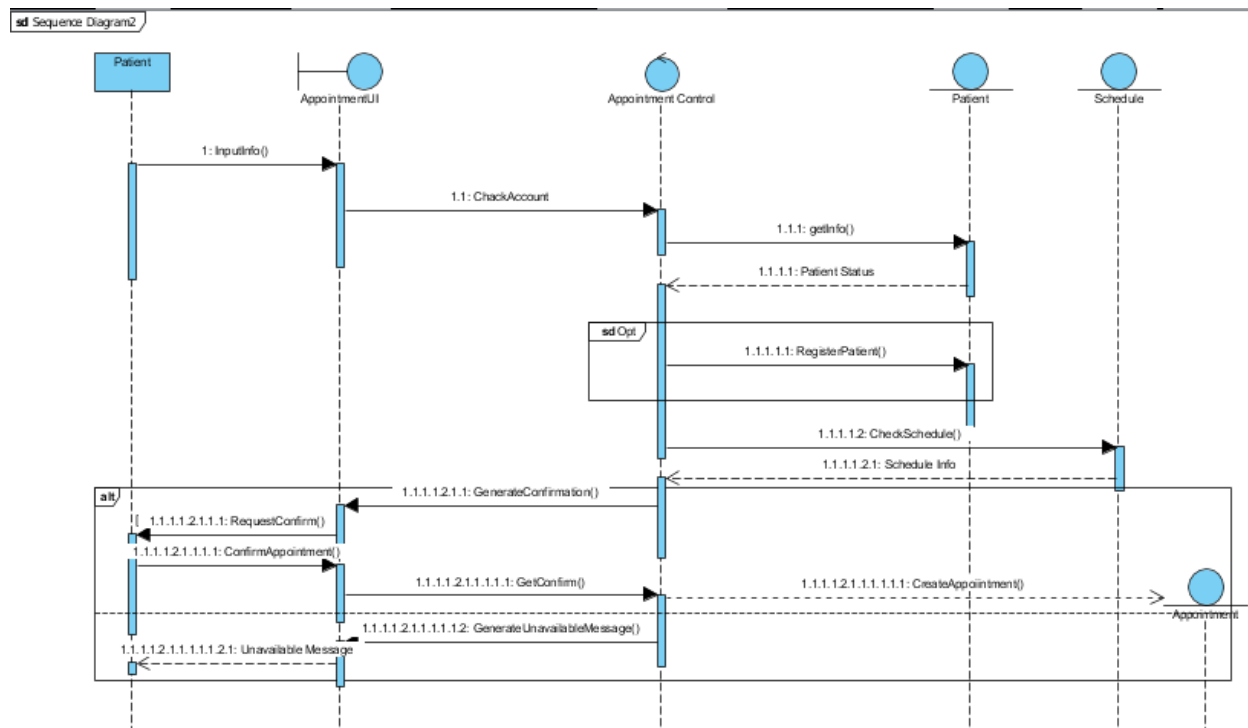
**** Almost same with question 9, student need to amend from answer of question 10

11. ABC clinic is a clinic that provides high-quality health care to residents of the area. The clinic provides services such as community health care, general medical consultation, treatment and so on.

When a patient wishes to make appointment with the doctor, the patient needs to provide some basic information such as patients' name and identity card number. The appointment system will check the patient's status. However, if the patient is a new patient, he / she must register first before they can make any appointment. For each appointment, the system will check the availability of the date and time of the doctor. If the doctor is available, the system will prompt for confirmation and add the patient's appointment details in the system. Otherwise, the patient is required to choose other slots for the appointment.

Construct a **design sequence diagram/collaboration diagram** that show the normal flow of events for the use case “Add patient's new appointment” for the online clinic appointment management system.

You are required to include boundary (UI), control, entity objects (account and customer) and object creation in your diagram.



12. Construct an **design sequence diagram/collaboration diagram** for the “Reserve bus tickets” use case for Online Bus Tickets Reservation System as described below:

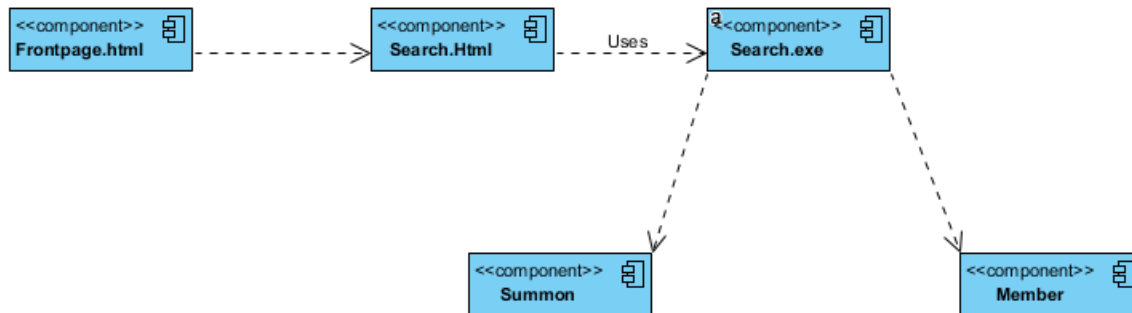
The customer is allowed to place bus tickets reservation based on the bus schedule availability via online. If the bus schedule is available, the system will check the seat availability. If the seat is available, the system will request customer’s personal details. The system will check the customer’s status. If he or she is a new customer, the system will register the customer details into the system; else the customer can proceed to make the reservation.

Once the reservation transaction is done, the system will display the total amount payable and the customer should make online payment one day before the departure date and time; else the system will automatically cancel the reservation. Once the payment transaction is done, the system should be able to generate the online ticket receipt and the customer is required to print out the receipt as a proof.

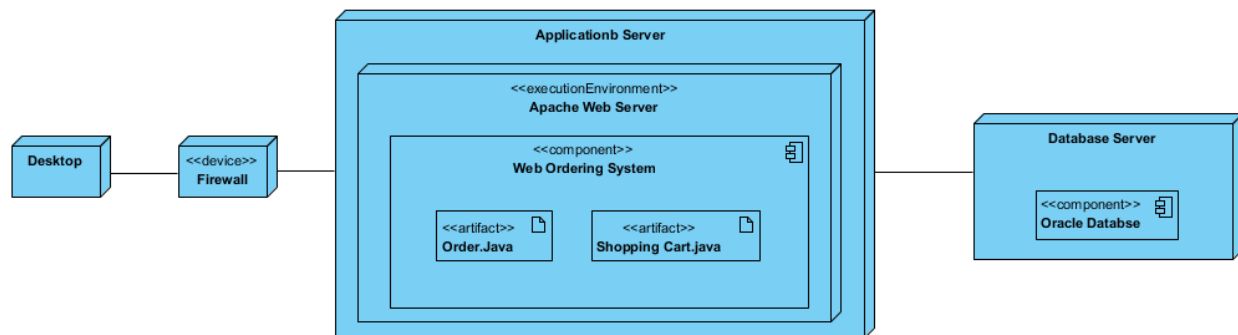
You are required to include boundary (UI), control, entity objects (account and customer) and object creation in your diagram.

****Quite similar with question 11, please try this question.

13. A country recently would like to implement an online Automated Enforcement System (AES) which include services such as check and pay summon. The system will help the users to check their five years summon records. If summon is not paid within a year, the user will be issued a letter to attend a court defend. Once summon is paid, it will be checked as "Cleared". The summon system consists of several components: frontPage.html, search.html, search.exe, and two tables: summon.tbl, and member.tbl. Draw a component diagram for the online summon system.



14. Draw a deployment diagram for web-based customer ordering system with the following nodes and artifacts: clients' desktop, application server, database server, firewall, ordering.java and shopping cart.java. You may make any relevant assumptions to support your diagram.



15. MyWarehouse.my is an online ordering system which collaborates with suppliers from different countries such as United Kingdom (UK), United States (US), Canada, New Zealand and Australia. MyWarehouse.my negotiates best price offered by suppliers and sell it through their online portal for a period of time with deadline. The customer can place order within the offered time. Once the customer has placed the order, they can check the order status from their account. MyWarehouse.my will check the payment and product availability before approve the order. After the order is approve, it will be sent to the associating overseas supplier. The order status will be changed from "Approved" to "Pending".

Once the deadline is expired, MyWarehouse.my will collect all the orders from the customer and inform the suppliers on the quantity to deliver. All products will be shipped directly from overseas supplier to MyWarehouse.my office in bulk. The order status will then be changed to “Shipped”. Once the products reach the office, MyWarehouse.my will then dispatch it to individual customer through local courier service. The customer can see “Dispatched” status in their account.

- (a) Construct a deployment diagram for the online ordering system. You should include nodes, artifact and communication path in your diagram.

Similar to question 14, need to make amendment.

- (b) Propose a simple three layered architecture for MyWarehouse.my online ordering system (package diagram).

